

SMART INDIA HACKATHON 2026

Judge / viva voce crib · not the 6-slide portal PDF

MOLE Q&A

120 spoken answers · traps marked · rapid-fire card

Team WE COOK · Problem SIH26025

Ministry of Coal · Hardware · Disaster Management

Speak the short answer first. Do not invent collapse %, ppm, or autonomy.

Tabletop demonstration · 9 September 2026

Spoken question-and-answer crib for institute screening and SIH judges

Field	Value
Event	Smart India Hackathon 2026
Official problem ID	SIH26025 (never write SIH2026025)
Problem title	Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India
Organisation / category / theme	Ministry of Coal · Hardware · Disaster Management
Team / idea	WE COOK · MOLE — Mine Observation & Live-alert Engine
Questions in this book	120 full answers · 24 marked TRAP · 20 rapid-fire lines
Public code	https://github.com/Rithvik-7/WE-COOK-MOLE
Document date	9 September 2026
Sister files	6-slide idea PDF (portal) · long judge document · this viva crib

How to use this book in the room

Read the question. Speak the answer in the navy box — 20 to 40 seconds. If the judge pushes, use the teal “If they push” line only when it exists. Red TRAP questions are where teams over-claim. Pause, then give the honest limit. Do not recast MOLE as a maze robot, a collapse-percentage app, or a certified mine instrument. Upload only the official 6-slide idea PDF on the SIH portal. This file stays with the team.

Honesty, in one paragraph

MOLE is a working tabletop demonstration. Two fixed nodes watch tilt, vibration, and (on Node A) a crack slider. The laptop runs rule-based early warning plus Isolation Forest and a 30-second sensor forecast. An officer can then drive a rover to look more closely. Readings can show disturbance on this model. They do not prove a collapse, name its cause, or certify that a real mine is safe.

Pocket card — always say / never say

Always say	Never say
Tabletop demonstration, not a certified mine	The mine is safe / certified / collapse imminent

Disturbance, not proven collapse or named cause	Collapse probability, risk score, or “87% chance”
Rules latch; ML cannot clear a red warning	Roof will fail in N minutes (as a geotech claim)
30 s sensor forecast — not a collapse prediction	The ground sank X centimetres from MPU tilt
Two radios: ESP-NOW/USB nodes, Wi-Fi AP rover	CO ppm from this MQ-7
Green = fresh valid data, not a safe mine	Autonomous navigation / auto-brake / lidar map

Remaining never-say lines:

- Camera vision AI
- LoRa / mesh / kilometre range (not built)
- SIH2026025
- Optional rover or decorative AI badge
- Node B is a surveyed geodetic reference
- Rover IMU is Node B
- Lives saved / disasters averted (invented)
- DGMS-approved instrument (this prototype is not)

Pocket card — rapid-fire facts

If a judge fires one-word questions, answer from this table. Thresholds are tabletop demo numbers, not DGMS trigger levels.

Cue	Answer in one breath
Official ID	SIH26025 (never SIH2026025)
Team / idea	WE COOK / MOLE — Mine Observation & Live-alert Engine
User	Safety officer on a tabletop model
Boards	ESP32 A, ESP32 B, Waveshare S3-Zero, ESP32 rover
Node radio	ESP-NOW ch 1 broadcast → USB JSON 115200
Rover radio	Wi-Fi AP Mine-Rover-AP, 192.168.4.1
WATCH	3° or 2 mm, 3 samples (tabletop)
ALERT	6° or 4 mm, 3 samples, latches
UNKNOWN	Stale > 5 s, bad IMU, A uncalibrated
Clear	3 fresh NORMAL; ack ≠ clear
AI	sklearn IF + LOF + joint IF + 30 s forecast
Prediction	Next 30 s of signals + MAE, not collapse %
Priors	Quiet tabletop, labelled Tabletop prior
Live train	≥120 NORMAL, live only, not in sim
IR	GPIO19 LOW=near; does not stop motors
MQ-7	Raw ADC, HIGH ≥ 2500, not ppm
ECHO	GPIO18 after 1k/2k divider
L298N	IN 13/12/14/27; GPIO19 is not IN4
Green	Fresh data, no tabletop trigger — not “mine safe”

Pitch order	Monitor → warn → inspect on command
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Contents

Section	Questions
1. Identity and problem statement	Q1–Q10 (10 questions)
2. Domain: subsidence, mines, and existing methods	Q11–Q20 (10 questions)
3. Solution, uniqueness, and what we refuse to claim	Q21–Q28 (8 questions)
4. Hardware — Node A, Node B, and the USB receiver	Q29–Q42 (14 questions)
5. Hardware — rover (core unit, separate radio)	Q43–Q52 (10 questions)
6. Communication, freshness, and failure modes	Q53–Q59 (7 questions)
7. Software, dashboard, and operator workflow	Q60–Q68 (9 questions)
8. Early-warning rules (the latch)	Q69–Q74 (6 questions)
9. Machine learning — the questions that win or lose “AI-enabled”	Q75–Q93 (19 questions)
10. Honesty traps — answer these slowly	Q94–Q103 (10 questions)
11. Feasibility, cost, scale, and field path	Q104–Q112 (9 questions)
12. Demo script, SIH process, and 36-hour questions	Q113–Q120 (8 questions)
13. Closing pitch sentence	Memorise this

1. Identity and problem statement

Start every judging round with the official ID, the user, and the honesty line. Do not recast the problem as a maze robot or a collapse-percentage app.

Q1 · What is your project, in one sentence?

MOLE is a tabletop mine-monitoring and inspection system: two fixed nodes watch tilt, vibration and a crack gap; the laptop latches an early warning and runs Isolation Forest plus a 30-second sensor forecast; then an officer can remotely drive a rover because a warning appeared. Readings show disturbance on this model. They do not prove a collapse.

Q2 · What is the official problem ID? Why not SIH2026025?

The catalogue ID is SIH26025 — Smart India Hackathon 2026, problem 025, Ministry of Coal, Hardware, Disaster Management. SIH2026025 is year-plus-025 concatenated. Judges, catalogues and the six-slide template match SIH26025. We never write the other string on the portal or the idea PPT.

Q3 · State the problem title.

Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India.

Q4 · Who is the user, and who owns the problem?

On demo day the user is a mine safety officer looking at a tabletop model. The Ministry of Coal owns the problem statement. We help that officer notice a concerning change, see the evidence, and inspect remotely. We do not replace a colliery control room or a DGMS-approved instrument.

Q5 · What does MOLE stand for? Team name?

Mine Observation & Live-alert Engine. Team name on the portal is WE COOK. Public code is github.com/Rithvik-7/WE-COOK-MOLE. Team ID is filled from the SIH portal after nomination.

Q6 · How does MOLE answer each word of the problem statement?

AI-enabled: named sklearn models, features, score and MAE on screen. Low-cost: ESP32-class boards we already have; no cloud bill. Real-time: about one summarised packet per second per node. Monitoring: two fixed nodes that keep watching. Prediction: next 30 seconds of tilt, vibration and crack millimetres, with MAE — not collapse percent. Early warning: WATCH then a latched ALERT from rules; ML cannot clear the latch. Underground coal mines: the PS context; the demonstration is a tabletop.

Q7 · Is ML optional? Is the rover optional?

Neither is optional. Hardware without visible ML does not answer “AI-enabled” or “prediction”. A rover without standing nodes is a maze car, not this PS. Pitch order is monitor and evidence first, then drive the rover because a warning appeared.

Q8 · TRAP · Is this a certified mine-safety system?

No. It is a working Smart India Hackathon Hardware demonstration on a tabletop model. Hobby ESP32 and MPU6050 boards are not intrinsically safe and are not DGMS-approved underground instruments. Green on our dashboard means fresh valid data and no configured tabletop trigger. It does not certify a mine.

Q9 · Give the 15-second architecture.

Node A and Node B, classic ESP32, ESP-NOW channel 1, into a Waveshare ESP32-S3-Zero that prints JSON on USB. Laptop Flask: store, rules, Isolation Forest, 30-second forecast, one local website. Rover is a separate classic ESP32 on Wi-Fi AP Mine-Rover-AP. Two radios on purpose.

Q10 · What must never appear on the six-slide portal PDF?

SIH2026025, collapse probability, lives saved, LoRa we did not build, autonomous driving, CO ppm from MQ-7, a seventh slide, a budget/BOM, or teammate photos. Upload only the official six-slide idea PDF. This Q&A book and the long judge document are for the room, not the portal.

2. Domain: subsidence, mines, and existing methods

Judges from mining or remote sensing will test whether you know the difference between a proxy and a geodetic measurement. Stay in that honesty. Context citations are not features we implemented.

Q11 · What is mine subsidence, in plain language?

Subsidence is downward or differential movement of the ground, often above underground workings, as the rock mass settles or is disturbed. It can show up as tilt, cracks, or unusual vibration at the surface or in the workings. Measuring it properly in a colliery uses surveyed monuments, approved geophones, extensometers, or satellite InSAR — not a student IMU on a model.

Q12 · TRAP · Does a tilt reading prove subsidence or a collapse?

No. Tilt is a change of orientation from a captured baseline. On our table it means the mount moved. It is not vertical settlement in centimetres, and it does not name the cause — blasting, a tap, a roof fall, or a vehicle. Measurements can indicate disturbance. They do not prove a collapse.

Q13 · How is this done in real Indian mines today?

Typical tools are periodic surveys and total stations, piezometers and extensometers, walk-in inspections, and at some longwall sites surface geophones. Satellite SBAS-InSAR is used in research and some monitoring programmes for wide-area deformation, but it is not a one-second heading for an officer at a face. DGMS (S&T) Technical Circular 01 of 2017, clause 6.1, asks longwall mines for surface geophones in strata-control plans. We cite that as field-grade context. We do not implement it.

If they push: If they ask for a paper: Sivakumar et al., 2005, on microseismic event-rate changes at Rajendra / SECL is the published Indian longwall hint we mention — again as context, not as our algorithm.

Q14 · Then why has a cheap IoT kit not replaced those methods?

Because underground coal is a hazardous atmosphere: equipment must be approved, mounts must be surveyed, radio must survive rock and regulation, and a false green light is dangerous. Cheap IMUs drift, ESP-NOW is unencrypted and short-range, and there are almost no labelled collapse events for a student model. The blocker is certification, labelling and environment — not that “nobody thought of ESP32”.

Q15 · Why is a tabletop model a valid answer to this Hardware PS?

SIH Hardware asks for a working loop a judge can verify in minutes: disturb → warn → explain → inspect. A tabletop lets us show live packets, a latch, named ML, and a rover without claiming a colliery installation. The field path — surveyed mounts, DGMS-approved sensing and communications, mine thresholds — is spoken as future work, not as today’s output.

Q16 · MPU6050 versus a geophone or an inclinometer — why this sensor?

MPU6050 gives gravity-based roll/pitch and a crude RMS vibration proxy on a 3.3 V I²C chip we can wire today. A geophone is a velocity sensor for seismic/microseismic arrays. A surveying inclinometer or total station measures displacement with a known datum. We are an honest low-cost proxy for mounting-surface micro-movement, not a professional microseismic array.

Q17 · TRAP · Will you quote CIL fatality numbers as impact?

Only as published ministry context if a judge asks why the PS exists — for example MoC Annual Report figures on accidents, subject to DGMS reconciliation. We do not invent “lives MOLE will save” or disasters averted. Impact we can defend is notice, evidence, and a remote look on the table.

Q18 · What does Coal Mines Regulations, 2017 have to do with this?

CMR 2017 sets duties around dangerous occurrences and strata control. It explains why an officer needs timely evidence. It is not implemented as software, and our 3° / 6° bands are not regulation trigger levels.

Q19 · Could InSAR replace your nodes?

InSAR is excellent for wide-area, millimetre-to-centimetre deformation over days to weeks. It does not give a one-hertz heading inside a model heading, and it does not drive a rover. We would treat InSAR as a later fusion path, not as something we built.

Q20 · Why two points on the surface, not a dense mesh?

Two points are enough to show local versus common motion on a table: Node A moves, Node B stays quiet. A real site would need a surveyed array. We do not pretend six hobby boards are a mine network.

3. Solution, uniqueness, and what we refuse to claim

If a judge has seen other student kits, this is the differentiation section.

Q21 · Walk through how it works, in order.

Nodes sample the MPU about 80 times a second and send one summary packet per second on ESP-NOW channel 1. The S3-Zero prints one JSON line per packet on USB at 115200. The laptop checks freshness and validity, converts tilt from the officer's baseline, converts Node A ADC to millimetres after two-point calibration, latches WATCH/ALERT from rules, scores Isolation Forest and a joint A–B forest, and forecasts the next 30 seconds of those same signals. The officer reads both columns. If they decide to look, they join Mine-Rover-AP and hold-to-move the rover.

Q22 · What is actually unique — not slogan unique?

One product, two radios, so rover vibration cannot poison Node B or node ML. Rules and ML sit side by side; Isolation Forest cannot talk a red warning back to green. Prediction is labelled as a 30-second sensor-trend forecast with MAE, not a collapse prediction. Node B is a comparison location, not a fake surveyed monument. Simulation is permanently labelled and cannot persist-train. Live USB never silently fakes data. Green does not mean certified safe.

Q23 · Why not one node?

One node cannot tell local disturbance from the whole table being bumped. Joint Isolation Forest uses absolute A–B tilt and vibration residuals and patterns such as QUIET, LOCAL_A, LOCAL_B, COMMON. That is still not geology. It is a localisation hint a judge can see.

Q24 · Why is Node B not a “reference station”?

A geodetic reference needs a surveyed stable monument. Node B is another hobby IMU on the model. We call it a comparison location. Calling it a reference would over-claim the physics.

Q25 · Why is intelligence on the laptop, not on the ESP32?

ESP32 measures and sends. Calibration, millimetres, A versus B, SQLite, sklearn, and the website need CPU, storage and a screen the officer already has. Firmware does not compute millimetres, alerts or AI. That split keeps the nodes simple and the claims auditable in Python.

Q26 · Why not pitch this as a gas-sensor maze robot?

That is an older, superseded identity. SIH26025 is monitoring, prediction and early warning of subsidence-style disturbance, then inspection. The rover exists because a warning appeared. MQ-7 is an inspection ADC on the rover, not the product, and it is not calibrated ppm.

Q27 · Walk-in inspection already exists. Why a rover?

Walk-in sends a person in first. Fixed nodes keep watching while the rover, if used, looks in a different place. On the table the officer inspects the model tunnel without putting a hand in first. It is remote drive, not autonomous rescue.

Q28 · TRAP · What do you explicitly refuse to claim?

Mine is safe or certified. Collapse or subsidence probability. Roof will fail in N minutes. Ground sank X centimetres. CO ppm from MQ-7. IR as auto-brake or lidar. Autonomous navigation. Camera vision. LLM geology. LoRa or colliery mesh. Login or cloud operations centre. ML clearing a latched alert.

4. Hardware — Node A, Node B, and the USB receiver

Pins and boards are frozen. If you invent a different MCU or GPIO in the room, you lose the as-built contract.

Q29 · List every board and its job.

Node A: classic ESP32-WROOM-32 DevKit — tilt, vibration, crack slider. Node B: same DevKit — comparison tilt and vibration, no slider. Receiver: WaveShare ESP32-S3-Zero (ESP32-S3FH4R2) — ESP-NOW to USB-C JSON, no sensors. Rover: classic ESP32-WROOM-32 — drive and inspect on its own Wi-Fi AP. That is four boards. The S3-Zero is infrastructure, not a third monitoring node.

Q30 · Node A pins and defines?

MPU6050 I²C address 0x68, SDA GPIO21, SCL GPIO22, 3.3 V, AD0 to GND. 10 kΩ linear slider SIG through 1 kΩ into GPIO34 (ADC1). NODE_ID 1, HAS_POT 1. Never 5 V into the ADC. Sketch: firmware/node/node.ino. Arduino board: ESP32 Dev Module.

Q31 · Node B — what is different?

Same MPU pins. Slider not fitted. sliderRaw is always -1. NODE_ID 2, HAS_POT 0. Healthy valid mask is 5 (IMU + vibration), not 7. Sketch: firmware/node_b/node.ino.

Q32 · How is tilt computed?

Firmware sends gravity roll and pitch: $\text{roll} = \text{atan2}(a_y, a_z)$, $\text{pitch} = \text{atan2}(-a_x, \sqrt{a_y^2 + a_z^2})$. That is trustworthy only when the mount is approximately still. The laptop, after the officer captures a baseline, computes tilt change as $\sqrt{(\text{roll} - \text{roll}_0)^2 + (\text{pitch} - \text{pitch}_0)^2}$. That is orientation change, not vertical ground displacement.

Q33 · What is vibration_g?

RMS residual of the three acceleration axes around their one-second mean, computed in firmware and sent as a proxy. It is not a calibrated seismic or geophone measure. A tap and a sustained disturbance look different in duration; we still do not classify blasting versus roof fall.

Q34 · How does the crack slider become millimetres?

Firmware sends raw 12-bit ADC only. The laptop applies a two-point ruler calibration: two distinct ADC values and two millimetre readings. Node A cannot show NORMAL until that scale exists. The linkage must be guided, not a free knob. Changing calibration resets Node A anomaly features.

Q35 · Why GPIO34 and a 1 kΩ series resistor?

GPIO34 is ADC1. ADC2 is not used with Wi-Fi or ESP-NOW. The 1 kΩ series resistor protects the pin. Optional 100 nF from GPIO34 to GND. Never feed 5 V into the ESP32 ADC.

Q36 · Sample rate and radio packet rate?

MPU is sampled about 80 Hz (SAMPLE_US 12500). Nodes broadcast about one summarised packet per second to FF:FF:FF:FF:FF:FF on ESP-NOW channel 1. Sequence increments after each send.

Q37 · Explain the valid bitmask.

IMU = 1, potentiometer = 2, vibration window = 4. Healthy Node A = 7, healthy Node B = 5. Invalid fields in USB JSON are null, never 0, so a dead sensor cannot look like a perfect zero reading.

Q38 · What does node firmware refuse to compute?

Millimetres, WATCH/ALERT, Isolation Forest, forecasts. The ESP32 measures and sends. That is deliberate so a judge can audit rules and ML in Python.

Q39 · Why Waveshare ESP32-S3-Zero, not a classic ESP32 or an S2?

We need native USB-C CDC at 115200, no CH340, no sensors on the gateway. The freeze is Waveshare ESP32-S3-Zero. Arduino: Waveshare ESP32-S3-Zero or ESP32S3 Dev Module, USB CDC On Boot = Enabled. It is not the Node A/B board and not ESP32-S2.

Q40 · What does the receiver do, and what does it not do?

Wi-Fi STA forced to channel 1, ESP-NOW receive, size/version/node_id check, one JSON object per line on Serial 115200. It may add gateway_ms. It does not compute tilt-delta, millimetres, rules or AI. Sketch: firmware/receiver_s3/receiver_s3.ino.

Q41 · Upload is stuck on Connecting... What do you do?

One USB board at a time. Hold BOOT, tap RESET, release BOOT, then upload. Classic ESP32 uses the same BOOT/RESET trick. Close Serial Monitor before Python — only one program may own the COM port.

Q42 · Give a USB JSON example the judge can look for.

Node A: type telemetry, schema 1, node_id 1, valid 7, roll_deg, pitch_deg, vibration_g, adc_raw. Node B: node_id 2, valid 5, adc_raw null. node_id 1 and 2 are reserved. Rover is device_id rover over HTTP, not this USB schema.

5. Hardware — rover (core unit, separate radio)

Raise wheels before any motor demo. GPIO19 is IR. ECHO needs a divider.

Q43 · Why is the rover in the build at all?

Fixed nodes cannot move. After a warning, the officer may need to look down a heading. The rover is that inspection unit: remote FWD/REV/LEFT/RIGHT/STOP plus ultrasonic centimetres, an IR near-flag, MQ-7 raw ADC, and rover IMU. It does not replace Node A or B. While it drives, the nodes keep watching.

Q44 · Rover motor pins and stop behaviour?

L298N IN1=13, IN2=12, IN3=14, IN4=27. Leave ENA/ENB jumpers ON. STOP is all IN low. Firmware also stops if no repeated command arrives for about 400 ms. If a side runs backward, swap that motor's two wires. Do not invent new GPIOs.

Q45 · Drive bit map?

STOP 0000, FWD 1010, REV 0101, LEFT 0110, RIGHT 1001 on IN1 IN2 IN3 IN4. HTTP GET on the rover: /forward /backward /left /right /stop /telemetry /.

Q46 · TRAP · Why is GPIO19 IR and not a motor pin?

Older drawings that put GPIO19 on L298N IN4 are wrong. IR OUT is GPIO19, INPUT_PULLUP, active LOW means near, majority-of-3 debounce. It does not stop the motors. Treating IR as auto-brake would be autonomy.

Q47 · HC-SR04 wiring — why a divider?

TRIG is GPIO5. ECHO is 5 V. It goes through a 1 kΩ / 2 kΩ divider into GPIO18. Never wire ECHO straight into the ESP32. The dashboard shows hobby centimetres, not a lidar map.

Q48 · TRAP · What does MQ-7 actually report?

GPIO36, raw 12-bit ADC. Local HIGH if raw ≥ 2500 (MQ7_HIGH_RAW). Not carbon monoxide ppm, not methane, not a certified gas instrument. Uncalibrated ppm would mislead a judge.

Q49 · TRAP · Rover IMU — can it be Node B?

Never. Rover vibration is inspection-only, tagged device_id rover. Mixing a moving IMU into Node B or into node Isolation Forest would poison the comparison story. This is a frequent trap question.

Q50 · Why Wi-Fi AP instead of ESP-NOW for the rover?

Drive needs request/response HTTP from the laptop. Nodes need a quiet, low-rate sensor broadcast into USB. Mixing those paths would put rover traffic on the monitoring radio and invite feature leakage. SSID Mine-Rover-AP, IP 192.168.4.1, open AP in firmware. Laptop must join that AP to drive. S3-Zero USB does not carry rover commands or video.

Q51 · TRAP · Is driving autonomous? Is there a camera?

No and no. Remote hold-to-move only. No camera, no vision model, no auto-brake, no SLAM. A camera feed would not be “AI vision” unless a vision model were actually implemented — and it is not in this freeze.

Q52 · IR and ultrasonic disagree. Is that a bug?

Not necessarily. IR is a digital near-flag, not range. In simulation, IR can trip under about 12 cm while ultrasonic “tight” is 25 cm, so they can disagree on purpose. On live hardware the pins are independent. Dashboard explains agree_near / ir_only / sonar_only / clear. IR still does not cut motors.

6. Communication, freshness, and failure modes

Judges will unplug USB or ask what happens if Wi-Fi dies. Have the UNKNOWN / STOP answers ready.

Q53 · Why ESP-NOW channel 1, broadcast, unencrypted?

ESP-NOW and the receiver's Wi-Fi STA must share a channel; we force channel 1. Broadcast to FF:FF:FF:FF:FF:FF avoids a pairing ceremony for two nodes and one gateway on a table. It is unencrypted and short-range. We do not claim LoRa, mesh, leaky feeder, or colliery networking.

Q54 · TRAP · What happens if USB dies in live mode?

Live never falls back to simulation. The service retries the serial port every 2 seconds. Nodes become UNKNOWN after 5 seconds without a valid packet. UNKNOWN is not green. We would rather show "we do not know" than a fake healthy mine.

Q55 · What happens if rover Wi-Fi dies?

Firmware stops motors after about 400 ms with no repeated command. The laptop also sends STOP twice on release or Space to beat in-flight HTTP races. Hold-to-move is the operator contract.

Q56 · Why 5 seconds for UNKNOWN?

Packets are about 1 Hz. Five seconds of silence is long enough to reject a glitch and short enough that a dead node cannot sit on a stale NORMAL. Freshness is laptop receipt time, not a fake clock on the node.

Q57 · Duplicate packets?

Schema 1, known node_id, non-decreasing sequence. Duplicates do not refresh freshness. Junk JSON is dropped.

Q58 · Why can the phone app not replace the laptop website?

The S3-Zero arrives over USB serial. A phone cannot see that COM port. Flask on 127.0.0.1:5000 is the operator app. No accounts, no venue cloud, no npm. Judges look at the laptop browser so nodes and rover stay one product.

Q59 · Do we need venue internet?

No. Nodes work over USB with Wi-Fi off. Rover needs only the rover AP. sklearn runs locally. Optional helper chatbot Pip may use a cloud key; it is not the product AI and the demo does not depend on it.

7. Software, dashboard, and operator workflow

Two pages on purpose. Drive controls must not appear on monitoring.

Q60 · What is the software stack?

Python 3, Flask, SQLite, numpy, scikit-learn, joblib, pyserial, requests, pytest. Charts are canvas, not a CDN Chart.js. Bind 127.0.0.1:5000. No login. From software/: pip install -r requirements.txt, then python app.py -mode simulate or --mode live --serial auto.

Q61 · Why two web pages?

Monitoring + AI is /monitoring (also /). Rover Inspection is /rover. If drive buttons sit on the monitoring page, judges treat MOLE as a gamepad. The story is evidence first, inspection second.

Q62 · Simulate versus live — rules a judge will test?

Simulate is permanently labelled Rehearsal. It cannot persist-train Isolation Forest (POST /api/train blocked). Tabletop prior .joblib files still load so IF can be READY; the UI says Tabletop prior. Live never silently fakes data. CSV export is tagged live or sim.

Q63 · How do you capture baseline and slider calibration?

Keep mounts still. Baseline A, then Baseline B. For the slider, record ADC+mm at two ruler positions with distinct ADC values and save two-point calibration. Node A stays UNKNOWN until that exists. Live baselines persist in SQLite across restarts.

Q64 · Ack versus clear — what does the officer actually do?

Ack records awareness. It does not erase history and does not un-latch ALERT. Clear is allowed only after three consecutive recovered NORMAL samples; otherwise the API returns 409. History still lists the event.

Q65 · TRAP · What is Pip?

An optional local helper chatbot with a handbook and FAQ, Mistral or Ollama if a key exists. Pip is not the SIH “AI-enabled” claim. Product AI is sklearn. If a judge points at a chat box, say that sentence first.

Q66 · Name the important Python modules in one breath.

app.py Flask and threads; engine.py ingest and snapshot; rules.py latch; anomaly.py Isolation Forest and LOF; forecast.py holdout regression; features.py vectors; calibration.py ADC to mm; rover_client.py HTTP to 192.168.4.1; serial_reader.py COM JSON; simulate.py rehearsal.

Q67 · How do you know the software contract still holds?

python -m pytest -q from software/. Contract tests cover simulate-cannot-train, AI cannot clear a latch, rover packet cannot update Node B, IR does not stop motors, and honesty on collapse % / ppm.

Q68 · Rising trend — what is it?

A simulate scenario: Node A climbs, Node B stays quieter. It is the judge-friendly forecast and localisation demo. Live equivalent is physically tilt Node A and leave Node B still.

8. Early-warning rules (the latch)

Say “tabletop demo numbers, not DGMS trigger levels” every time thresholds come up.

Q69 · State the WATCH and ALERT rules.

WATCH: tilt $\geq 3^\circ$ or crack ≥ 2 mm, three consecutive samples. ALERT: tilt $\geq 6^\circ$ or crack ≥ 4 mm, three consecutive samples. ALERT latches. Files: software/rules.py.

Q70 · Why three consecutive samples?

A single spike can be a knock or ADC glitch. Three samples at about 1 Hz is a short persistence check a judge can watch. It is not a geotechnical dwell time.

Q71 · Why latch? Why can Isolation Forest not clear it?

Early warning must not flicker off because a model score improved. The officer acknowledges, recovers the mounts, and clears only after three fresh NORMAL samples. ML may add UNUSUAL. It may not hide a rule.

Q72 · When is a node UNKNOWN?

Stale more than 5 s, unseen node, invalid IMU, or Node A slider not millimetre-calibrated. UNKNOWN is not NORMAL. Silence is not safety.

Q73 · Can Node B cancel Node A?

No. Comparison never vetoes the disturbed point. That would be a dangerous design even on a tabletop.

Q74 · TRAP · Are 3° and 6° mine standards?

No. They are visible tabletop bands so a judge can tilt a board and see WATCH then ALERT. Real sites would need surveyed mounts and approved trigger policy. We will say that even if they press for a standard.

9. Machine learning — the questions that win or lose “AI-enabled”

Name the models. Point at features, train count or Tabletop prior, score, LOF vote, MAE. Never say collapse probability. Isolation Forest is real AI; the PS word prediction is the 30-second forecast.

Q75 · What is the AI in this product?

Laptop scikit-learn only. Per node: Isolation Forest (contamination 0.05) plus Local Outlier Factor, with StandardScaler. Joint Isolation Forest on |A-B| tilt and vibration. Holdout-selected regression forecasts the next 30 seconds of tilt, vibration and Node A gap, with MAE. Explanations are number templates, not an LLM. Rover IMU, IR, ultrasonic and MQ-7 are not ML features.

Q76 · Explain Isolation Forest so a non-ML judge gets it.

Liu, Ting and Zhou, IEEE ICDM 2008. The algorithm grows random trees that isolate points. Unusual combinations of our features take fewer splits to isolate than the quiet cloud. We train on this rig's NORMAL behaviour. Output is UNUSUAL versus this tabletop, plus a score and the top standardised feature — not a collapse probability.

Q77 · Why Isolation Forest, not a neural net or SVM?

We have unlabelled quiet data, not a library of mine collapses. Isolation Forest is unsupervised, small-data friendly, inspectable, and standard in sklearn. A neural net would overclaim and overfit thirty noisy IMU points. Judges can see feature names and a score. That is the point.

Q78 · What does contamination 0.05 mean here?

It is sklearn's expected outlier fraction inside the forest — a sensitivity knob, not a scientific statement that 5% of a mine is collapsing. We train only on samples rules already called NORMAL.

Q79 · What features go into the per-node forest?

tilt_change_deg, vibration_g, d_tilt (one-second rate), and Node A relative_mm once calibrated. Node B has no millimetre feature. Features are scaled. The officer sees the names and which one deviated most.

Q80 · Why add Local Outlier Factor?

LOF compares local density to neighbours. Isolation Forest isolates globally with random partitions. Together they vote. The dashboard shows Isolation Forest state, LOF state, and vote count. LOF is support, not a second product.

Q81 · What is the joint forest?

A third Isolation Forest on [abs(A_tilt-B_tilt), abs(A_vib-B_vib)]. Patterns such as QUIET, LOCAL_A, LOCAL_B, COMMON. It is a localisation hint: did one mount move or did the whole model move? It is not a geology or fault model.

Q82 · INACTIVE / READY / UNUSUAL?

INACTIVE: no usable model. READY: model loaded, current vector not flagged unusual. UNUSUAL: forest and/or LOF flag this combination versus trained normal. Tabletop priors ship READY without waiting 120 live samples; the label is Tabletop prior, not “trained on a real mine”.

Q83 · When are you allowed to retrain live?

Live mode only, at least 120 NORMAL samples per node, blocked if WATCH, ALERT, UNKNOWN or simulate. Writes nodeN.joblib and overrides the prior until reset. Simulate cannot persist-train so a rehearsal cannot pretend to be a field model.

Q84 · TRAP · Are the shipped priors a real-mine model?

No. artifacts/node1.prior.joblib, node2.prior.joblib and joint.prior.joblib describe synthetic quiet-tabletop NORMAL so a demo can start READY. After a physical baseline, scores may be noisy until optional live retrain on that MPU.

Q85 · TRAP · The PS says prediction. What did you actually build?

Sensor-trend prediction: last roughly 30–60 one-hertz points; candidates Ridge, HuberRegressor, LinearRegression versus a persistence baseline; holdout MAE picks the winner; horizon is the next 30 samples of the same signals. On-screen label: 30 s sensor trend forecast — not a collapse prediction. Mine-subsidence or collapse prediction is not claimed. That needs surveyed mounts, labelled events and geotech validation.

Q86 · Why 30 seconds, not 30 minutes or 30 days?

The officer is watching a live heading. Thirty seconds is long enough to see a dashed line and check MAE, short enough that a linear model on IMU noise is honest. Multi-day subsidence is InSAR and survey territory. We will not put a 30-day roof forecast on the screen.

Q87 · Why holdout MAE instead of always using Ridge?

Quiet data can be almost flat; LinearRegression or persistence may win with tiny MAE. Huber is robust to spikes but may fail to converge — it is a candidate, not sacred. Holdout MAE is the evidence the officer sees. We do not hide a worse model behind a brand name.

Q88 · What is the persistence baseline?

The naive forecast “next value equals last value”. If Ridge cannot beat that on holdout MAE, we should not brag. Showing the competition is part of not doing decorative AI.

Q89 · TRAP · Time-to-WATCH / time-to-ALERT — is that a collapse clock?

Only if the fitted slope heads toward the tabletop threshold. Otherwise it is null. Even then it is “if this rate holds on these signals”, not “roof fails in N minutes”. Quiet or falling slope must not invent a countdown.

Q90 · MAE is nearly zero. Are you overfitting?

On a flat quiet signal, predicting “stays flat” has MAE near 0. That is a boring truth, not magic. Say that sentence. Do not call it 99% accuracy.

Q91 · Why not LSTM / Prophet / ChatGPT for prediction?

Thirty noisy points, no labelled disasters, offline laptop, inspectable MAE. An LLM inventing geology is exactly the AI slop this PS should not reward. Pip, if shown, is a helper for operators, not the mine model.

Q92 · Can UNUSUAL exist while rules are still NORMAL?

Yes. That is the point of the forest: unusual combination below the 3° latch. The officer sees both columns. Conversely, a latched ALERT stays even if the forest later looks quiet.

Q93 · n_estimators, random_state — any numbers to remember?

IsolationForest n_estimators=100, contamination=0.05, random_state=0, plus StandardScaler. MIN_TRAIN live = 120 samples. Forecast HORIZON = 30, lag features from the last 6 samples plus mean and delta.

10. Honesty traps — answer these slowly

If you only memorise one section, memorise this one. Wrong answers here sink an otherwise working demo.

Q94 · TRAP · Green / NORMAL — is the mine safe?

NORMAL means fresh valid packets and no configured tabletop trigger. It does not certify a mine, a roof, or a workplace as safe. Hobby electronics are not approved for an explosive atmosphere.

Q95 · TRAP · Give me the probability the roof will collapse.

We do not produce collapse or subsidence probability. Isolation Forest scores unusual versus this rig's normal. The forecast is the next 30 seconds of measured signals with MAE. Those are different claims.

Q96 · TRAP · The ground sank 10 cm. Can you show that?

No. Tilt is orientation change from baseline. The slider is a model gap in millimetres after ruler calibration. Neither is vertical geodetic settlement.

Q97 · TRAP · What is the CO reading in ppm?

We do not quote ppm from this MQ-7. You will see raw ADC and a local NORMAL/HIGH flag at threshold 2500.

Q98 · TRAP · Does IR stop the rover automatically?

No. IR is a flag. Motors stop when the officer releases, hits Space, or the 400 ms watchdog fires. Auto-stop on IR would be autonomy we did not build and did not test as a safety function.

Q99 · TRAP · You used LoRa / 4G / a mine leaky feeder, right?

No. Nodes are ESP-NOW tabletop into USB. Rover is a local Wi-Fi AP. Field radios are future work after an RF and approvals survey.

Q100 · TRAP · Show me the AI camera.

There is no camera in this freeze. Inspection is centimetres, IR flag, MQ-7 raw, rover IMU. A JPEG on a phone is not computer vision.

Q101 · TRAP · Your chatbot is the AI, yes?

No. sklearn on the laptop is the product AI. A helper named Pip answers from a handbook. If the chat box is off, Isolation Forest and the forecast still run.

Q102 · TRAP · If USB fails, do you keep showing the last nice chart as live?

No. Nodes go UNKNOWN. Live mode does not switch to simulation. Simulation, when used, is bannered.

Q103 · TRAP · How many lives will this save?

We do not know and we will not invent a number. Benefit a judge can verify: see A versus B, see why a warning latched, see model names and MAE, inspect without entering the model tunnel first, keep a CSV history.

11. Feasibility, cost, scale, and field path

SIH scoring cares about feasibility. Be concrete about what is in the repo versus what still needs a soldering iron.

Q104 · Can you actually demo this week?

Firmware sketches, Flask site, rules, IF+LOF+joint, holdout forecast, priors, simulate scenarios and rover proxy are in the repository. Hardware day still requires: wire the freeze (especially echo divider and GPIO19 = IR), flash four boards, prove JSON on the S3-Zero at 115200, live baseline and slider cal, join Mine-Rover-AP, polarity test, STOP watchdog. SIH26025 is Hardware — extra sklearn does not replace working boards.

Q105 · You already have the parts. What did “low cost” mean?

ESP32-class COTS, no cloud bill, no licence server. We do not put a ₹ BOM on the idea PPT and we do not block the story on an old ₹4,000 node-only figure. Low-cost is the architecture, not a shopping-channel claim.

Q106 · How would this scale to a real mine?

More surveyed nodes, site-calibrated thresholds, approved sensing and comms (possibly leaky feeder, mesh, or mine LTE — after RF survey), integration with existing strata monitoring, labelled events before any collapse model, intrinsically safe enclosures. That is a multi-year instrumentation project. Today is the tabletop loop that proves the software split: measure on the edge, decide and learn on a local computer, inspect on command.

Q107 · Power and battery?

Demo is USB or bench supplies and a rover pack. We do not claim shift-long intrinsically safe batteries. If they ask field power: that is part of the approved-equipment path, not this prototype.

Q108 · Range through rock?

Not demonstrated. ESP-NOW and a 2.4 GHz AP across a table are not a colliery RF design. We will not quote a kilometre figure.

Q109 · Cybersecurity? Open AP?

Tabletop only. ESP-NOW is unencrypted. Rover AP has no password in firmware so a demo laptop can join without a sticky note. A field system would need authenticated radios and a closed network. We do not pretend the prototype is hardened OT.

Q110 · Main risks, spoken?

Over-claiming tilt as settlement. Mixing rover IMU into nodes. False comfort from a dead sensor — mitigated by UNKNOWN. Uncalibrated slider — Node A cannot go NORMAL. IR/ultrasonic disagreement — explained, no auto-brake. Hobby electronics underground — out of scope.

Q111 · Intellectual property / open source?

Implementation is in the public WECOOK-MOLE repository for the hackathon. Isolation Forest is a published algorithm; sklearn is BSD-licensed. We are not selling a certified product in this round.

Q112 · What is explicitly leftover?

Physical flashing, COM port on the venue laptop, ESP-NOW reliability across the table, slider mechanics, motor polarity, echo divider on the real board, IR aiming, AP join, live baseline, optional live retrain. Camera, LoRa, ppm, autonomy, collapse % remain out of scope unless the freeze is formally changed.

12. Demo script, SIH process, and 36-hour questions

Internal college rounds often ask “what have you built so far?” and “what will you do in 36 hours?” Answer with artifacts, not hope.

Q113 · Show me the judge demo, step by step.

1) Both nodes updating; say SIH26025: monitor, warn, inspect. 2) Point at Isolation Forest, prior or train count, features, forecast MAE; say not collapse percent. 3) Rising trend or tilt Node A; keep B quieter. 4) WATCH then ALERT with the measured reason; rules and ML side by side. 5) Open Rover Inspection because a warning appeared. 6) Wheels up; hold FWD; STOP; centimetres, IR, MQ-7 raw; IR does not auto-brake. 7) Ack; recover; clear after three NORMAL; history remains.

Q114 · Hardware fails during judging. What then?

`python app.py --mode simulate`, Rehearsal banner visible, same two pages, Rising trend. Say clearly it is labelled simulation. Do not pass sim off as USB. Isolation Forest still loads tabletop priors; persist-train stays blocked.

Q115 · What is already built versus what 36 hours would add?

Already built: firmware, local website, ML, tests, idea PPT. A grand-finale 36 hours is integration and reliability — flash, mounts, calibration, radio, rover polarity, a quiet 2-minute live retrain if time — not a new architecture. We will not promise LoRa or a camera overnight.

Q116 · Why this team of six?

Split by unit, not “everyone codes”: Node A mount and slider; Node B and table mechanics; S3-Zero radio and USB; rover mechanics and payload; laptop ingest/rules/rover proxy; dashboard, ML evidence, pitch. There is no fourth monitoring node for the sixth person.

Q117 · What goes on the SIH portal?

PDF of the official six slides only. ID SIH26025. This Q&A PDF and the long judge document stay with the team and the SPOC. They are not a seventh slide.

Q118 · Spoken close — the sentence to end on?

Isolation Forest and LOF flag unusual combinations versus this rig’s learned normal. A joint forest watches A versus B residual. Holdout-selected sklearn regression forecasts the next 30 seconds of those same signals with MAE. Rules remain the independent early-warning latch. The rover is remote inspection, not autonomy.

Q119 · Name the references you actually use.

Liu, Ting, Zhou — Isolation Forest, ICDM 2008. scikit-learn Ridge / Huber / LinearRegression and MAE. InvenSense MPU-6000/6050 spec. Espressif ESP-NOW guide and ESP32 ADC1 notes. Waveshare ESP32-S3-Zero wiki. MQ-7 datasheet as raw ADC. GitHub WECOOK-MOLE. CMR 2017 and DGMS (S&T) Circ. 01/2017 cl. 6.1 as context only.

Q120 · If a mentor says add camera, LoRa, and collapse percent tonight?

We will not. Those contradict the hardware freeze and the honesty table. We can discuss them as a field roadmap. Shipping a fake percent would fail the PS more badly than a smaller honest demo.

13. Closing pitch sentence

Isolation Forest + LOF flag unusual combinations versus this rig's learned normal. A joint forest watches A vs B residual. Holdout-selected sklearn regression forecasts the next 30 seconds of those same signals with MAE. Rules remain the independent early-warning latch. The rover is remote inspection, not autonomy.

14. Pin freeze (if they ask a GPIO)

Unit	Frozen detail
Node A / B MPU	SDA 21, SCL 22, 0x68, 3.3 V
Node A slider	10 kΩ linear, SIG → 1 kΩ → GPIO34, NODE_ID 1 HAS_POT 1
Node B	No slider, NODE_ID 2 HAS_POT 0, sliderRaw = -1
Receiver	Waveshare ESP32-S3-Zero, USB CDC On Boot Enabled, 115200 JSON
Rover motors	L298N IN1–IN4 = 13 / 12 / 14 / 27; ENA/ENB jumpers ON
Rover inspect	TRIG 5, ECHO 18 via 1k/2k, MQ-7 GPIO36 raw, IR GPIO19 LOW=near
Rover radio	Mine-Rover-AP, 192.168.4.1, not on the node USB path

15. Related files

- Portal upload: pitch/SIH26025_WE_COOK_IDEA.pdf (six slides only).
- Long technical write-up: pitch/SIH26025_WE_COOK_MOLE_Judge_Document.pdf.
- This viva crib: pitch/SIH26025_WE_COOK_MOLE_Judge_QA.pdf.
- Speaker crib for the six slides: pitch/SPEAKER-CRIB.md.
- As-built contract: AGENTS.md. How to run: README.md.

Close: nodes detect → rules latch → Isolation Forest + Ridge explain → rover inspects. Team WE COOK · SIH26025 · tabletop only.